Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Continuous Process Improvement with FMEA

TGS Ref No: TGS-2021009031

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version v11.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v10.0	Historical reference	Legacy lesson structure	Tertiary Infotech Academy Pte Ltd
v11.0	13 August 2026	Aligned 2-day plan, ten activities and current 146-slide deck	Tertiary Infotech Academy Pte Ltd

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Course Information

Course	Continuous Process Improvement with FMEA
Course code	TGS-2021009031
TSC	Continuous Process Improvement (ELE-PIN-4001-1.1)
Duration	2 days / 16 hours
Training	13.5 hours
Assessment	2.5 hours: WA 60 minutes + PP 90 minutes
Method	Facilitated concepts, case discussion, demonstration, ten connected hands-on activities and reflective review

Learning Outcomes

- ELO1: Create a continuous process improvement system.
- ELO2: Perform root cause analysis for continuous process improvement.
- ELO3: Design and apply Failure Mode and Effects Analysis (FMEA).
- ELO4: Identify and direct improvement opportunities.
- ELO5: Manage and evaluate improvement with a Process Control Plan.

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Day 1 Schedule

Time	Duration	Topic / activity	Learning outcome and content	Facilitation / learner activity	Resources / slides	Evidence / check
09:00–10:30	1.5 h	Administration and course orientation	Attendance, introductions, outcomes, competency mapping, assessment briefing and Orion case.	Facilitator briefing; learner introduces a workplace process and improvement need.	Slides 1–17	Participation; assessment conditions understood
10:30–10:45	Included	Short break	Refreshment break within the scheduled session.	Return on time.	—	Attendance maintained
10:45–13:00	2.25 h	Topic 1 + Activities 01–02	Create a CI system; culture, governance, PDCA, SIPOC, customer/process voice and coaching.	Case analysis; complete CI charter and SIPOC/PDCA workbooks; peer scope review.	Slides 19–39; Slides 33–35; Slides 36–38	Charter, roles, governance, SIPOC and PDCA hypothesis
13:00–14:00	Not counted	Lunch	Lunch break.	—	—	—
14:00–15:30	1.5 h	Topic 2 concepts	Problem statements, cause/effect levels, Pareto, Fishbone, evidence-backed 5 Whys, Fault Tree and data quality.	Interpret a defect Pareto; distinguish symptom, mode, effect and cause; test causal statements.	Slides 40–60	Bounded investigation statement and candidate-cause tests
15:30–15:45	Included	Short break	Refreshment break within the scheduled session.	Return on time.	—	Attendance maintained
15:45–18:00	2.25 h	Activities 03–04 + Day 1 review	Prioritise the defect pattern and verify root causes using complementary tools.	Complete Pareto, stratification, Fishbone, 5 Whys, Fault Tree and verification matrix; explain one confirmed mechanism.	Slides 54–56; Slides 57–59	Reconciled analysis, verification evidence and Day 1 reflection

Day 1 total: 8.0 scheduled training hours

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Day 2 Schedule

Time	Duration	Topic / activity	Learning outcome and content	Facilitation / learner activity	Resources / slides	Evidence / check
09:00–10:30	1.5 h	Topic 3 concepts	FMEA purpose, types, seven-step method, failure chain, controls, S/O/D, RPN cautions, action priority and review triggers.	Construct and critique failure chains; compare equal-RPN risk profiles.	Slides 62–92	Coherent function–mode–effect–cause logic
10:30–10:45	Included	Short break	Refreshment break within the scheduled session.	Return on time.	—	Attendance maintained
10:45–12:30	1.75 h	Activities 05–06	Build, rate and optimise the Orion PFMEA.	Complete PFMEA rows, S/O/D rationale, RPN formulas, actions, ownership, verification and re-score.	Slides 86–88; Slides 89–91	Rated PFMEA and evidence-based residual-risk decision
12:30–13:30	Not counted	Lunch	Lunch break.	—	—	—
13:30–14:30	1.0 h	Topic 4 + Activities 07–08	Select, pilot and direct risk-reduction actions with transparent criteria and coaching safeguards.	Use weighted/Pugh comparison; draft a bounded pilot and coaching plan.	Slides 93–116; Slides 110–112; Slides 113–115	Recommendation and scale/revise/stop rule
14:30–15:30	1.0 h	Topic 5 + Activities 09–10	Variation, stability, capability, SPC signals, Control Plan, reaction plan and closed-loop review.	Interpret a control chart; calculate capability; complete traceable operational controls.	Slides 117–139; Slides 133–135; Slides 136–138	Evaluation statement and deployable Control Plan
15:30–16:30	1.0 h	Written Assessment (SAQ)	Four open-ended questions covering K1–K4.	Individual, open book under assessment conditions.	WA paper; LMS link	Completed WA responses
16:30–18:00	1.5 h	Practical Performance	Five connected tasks covering A1–A10.	Individual completion of assigned practical evidence; submit and sign the Assessment Summary Record.	PP paper; activity workbooks; LMS link	Completed practical evidence and signed record

Day 2 total: 5.5 training + 2.5 assessment = 8.0 hours

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Assessment Plan

Instrument	Duration	Coverage	Conditions	Evidence
Written Assessment (SAQ)	60 min	K1–K4	Individual; open book	Four written responses
Practical Performance	90 min	A1–A10	Individual; use supplied files	Five task outputs and assessor record

Facilitator Preparation

- Verify digital-attendance access and learner materials before class.
- Download the current deck, Learner Guide and all ten activity folders.
- Check Microsoft Excel can open, calculate and save the supplied files.
- Keep answer keys trainer-only and distribute only candidate papers at assessment.
- After every activity, review the stated acceptance criterion and evidence—not only template completion.

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